

LLM-Serving Scheduler Portability Benchmark (LSSP): Does Comparative Scheduler Ranking Generalize Across Workload Sources, Operating Regions, and Metrics?

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Abstract

Large language model (LLM) serving systems increasingly rely on sophisticated scheduling mechanisms, and prior work has separately established both that these mechanisms are numerous and mechanistically diverse, and that the workloads used to evaluate them are themselves heterogeneous across source, time, and operating region. We ask a second-order question that follows from combining these observations: is the *comparative ranking* a scheduler benchmark reports itself portable across independently sourced workloads, operating regions, and evaluation metrics? We formalize this as the portability of $R(W, M, L)$, a comparative scheduler ranking induced by a fixed policy panel under a fixed benchmark protocol, as W (workload source/window), M (metric), and L (operating/load region) vary. We present the LLM-Serving Scheduler Portability Benchmark (LSSP): a preregistered, paired, multi-source evaluation of 13 scheduling policies across three independently released workload sources, a calibrated six-region operating grid, and an explicit completion-conditioned metric contract. We establish, as evidence available now, that the three workload sources are measurably different and strongly separable in their observed descriptor distributions, and, as historical pilot evidence, that scheduler differentiation is real but strongly source-dependent. LSSP’s central ranking-portability, sample-complexity, and real-system-validation results are not yet available at the time of this manuscript and are stated as explicit, outcome-neutral result contracts rather than asserted in either direction.

Contents

1	Introduction	3
2	Related Work	5
2.1	Serving Architectures and Scheduling Mechanisms	5
2.2	Production Workloads and Characterization	6
2.3	Simulators and Benchmark Frameworks	6
2.4	Workload Dependence and Benchmark Sensitivity	6
2.5	Ranking Portability and Benchmark Methodology	7
2.6	Position of LSSP	7
3	Study Design	8
3.1	Research Questions	8

3.2	Study Overview	8
3.3	Evidence Independence / Publication Boundaries	8
3.4	Experimental Unit and Paired Design	9
3.5	Preregistration	9
4	Workload Corpus and Characterization	9
4.1	Sources	9
4.2	Representation and Provenance	10
4.3	Window Construction	10
4.4	Descriptors	10
4.5	Cross-Source Differences	10
4.6	Source Separability	11
4.7	Drivers of Differences	11
4.8	Window-Size Sensitivity	11
5	Scheduler Benchmark Methodology	11
5.1	Scheduler Panel	11
5.2	Fidelity Taxonomy	12
5.3	Execution Model	12
5.4	Stage-0 Discriminability Pilot (Historical)	12
5.5	Operating-Region Calibration	13
5.6	Metrics	13
5.7	Paired Design and Repetitions	14
5.8	Mechanism Telemetry	14
5.9	Robustness Analyses	14
6	Scheduler Ranking Portability	14
6.1	Discriminability	14
6.2	Cross-Source Rankings	15
6.3	Pairwise Reversals	15
6.4	Temporal/OOD Portability	15
6.5	Load Conditioning	15
6.6	Metric Conditioning	15
6.7	Robustness	15
7	Benchmark Sample Complexity	15
8	Real-System Validation	16
9	Discussion	16
9.1	If Ranking Portability Is High	16
9.2	If Ranking Portability Is Low	16
9.3	Mechanism Equivalence and Collapse	17
9.4	Workload Descriptor Interpretation	17
9.5	Benchmark Construction and Practical Reproducibility	17
10	Threats to Validity	17
11	Reproducibility and the LSSP Artifact	19

1 Introduction

Large language model (LLM) serving systems have grown increasingly sophisticated in how they schedule and allocate resources to concurrent requests. Beyond simple first-come-first-served execution, modern serving stacks employ continuous batching and paged key-value (KV) cache management [8], iteration-level scheduling [24], chunked prefill to balance prefill and decode work within a batch [1], disaggregated prefill/decode execution across dedicated hardware pools [9, 27], KV-cache-centric disaggregated architectures [10], dynamic instance-level rebalancing [17], explicit fairness objectives [16], output-length-aware admission and memory packing [7], and learning-to-rank decode-priority policies [6]. Each of these mechanisms was introduced, and evaluated, against a particular choice of workload: a specific trace, a specific arrival process, a specific mix of prompt and output lengths, and a specific operating load.

A separate line of work has documented that these workloads are themselves far from uniform. Public production traces from different services and time periods differ substantially in burstiness, request concurrency, prompt and output length distributions, and failure characteristics [9, 18, 21]; large-scale characterization studies of KV-cache reuse and of year-long production traffic report further axes of heterogeneity across clients, models, and time [3, 15]; and dedicated workload-generation frameworks exist precisely because no single fixed trace is thought to be representative of production traffic in general [23]. Individually, prior work has therefore already established two things: that LLM-serving mechanisms are numerous and mechanistically diverse, and that the workloads used to evaluate them are themselves heterogeneous and shift across source, time, and operating region.

What has not been directly asked is a second-order question that follows from combining these two observations. The typical evaluation question in this literature is: *how well does scheduler A perform, relative to scheduler B, on the workload(s) chosen for this paper’s experiments?* Given that workloads are heterogeneous, and that a paper’s experiments necessarily use only a finite, chosen set of workloads, a comparative claim of this form is only as useful as its *portability*: does the same comparative ordering of schedulers hold if the workload distribution, the operating load, or the evaluation metric is changed to another one a practitioner might reasonably care about? This is a different question from workload heterogeneity itself, and a different question from any single scheduler’s sensitivity to load or SLO. It is a question about the reliability of the *comparison*, not about any individual system.

We formalize this as the portability of a comparative scheduler ranking $R(W, M, L)$, where W denotes a workload distribution, window, or source; M denotes an evaluation metric; and L denotes an operating or load region. $R(W, M, L)$ is the ordering (full or partial) that a fixed panel of scheduling policies induces under a fixed benchmark protocol, holding W , M , and L at specified values. Ranking portability asks how R behaves as W , M , or L is varied: whether the ordering of policies is preserved, where and how it reverses, and how much evidence (how many independent workload windows, drawn from how many sources) is required before a reported ranking can be trusted to generalize beyond the exact windows used to produce it.

This question matters because a scheduler comparison is scientifically and practically useful only insofar as its comparative conclusion transfers to the workload distributions a reader actually cares about. A benchmark result of the form “scheduler A outperforms scheduler B ” is not a claim about A and B in the abstract; it is implicitly a claim about A and B *under the workload, metric, and load conditions the benchmark used*. If that comparative ordering is fragile under source, metric, or

load-region substitution, then benchmark results risk being read as more general than they are, and evaluation effort risks being spent on a single workload family when the resulting conclusion may not transfer. Conversely, if comparative orderings are portable across independent sources, metrics, and load regions, that is itself a useful, non-obvious finding: it would mean a leaner benchmark protocol is sufficient for future scheduler comparisons, and it would give scheduler designers a much stronger warrant for claims of general improvement.

We refer to the benchmark built around this question as the LLM-Serving Scheduler Portability Benchmark (LSSP). LSSP asks the following research questions, fixed under a preregistered protocol before any comparative outcome was generated (§3):

RQ1. How stable are scheduler rankings across independent workload sources?

RQ2. How stable are scheduler rankings under temporal, provider, and domain shifts?

RQ3. To what extent do rankings obtained on synthetic stress workloads transfer to rankings on independent real-trace-derived workloads?

RQ4. Which source-native observable workload characteristics are associated with cross-distribution scheduler rank reversals?

RQ5. How many independent workload windows are required before a comparative scheduler ranking becomes statistically stable?

RQ6. Do representative simulated scheduler rankings and cross-workload rank reversals reproduce on a real serving engine?

Load-level dependence is deliberately treated as a secondary robustness/sensitivity axis applied to RQ1–RQ4, rather than as a headline research question in its own right, to avoid duplicating a closely related prior load-scaling study on an overlapping window set (§3).

Contributions. Because LSSP’s comparative scheduler outcomes had not yet been generated at the time this manuscript was prepared (§3), we state our contributions in a form that does not depend on, and is not falsified by, either possible direction of that future result:

- We formulate the portability of a comparative scheduler ranking, $R(W, M, L)$, as a benchmark object in its own right, distinct from both workload characterization and single-scheduler evaluation.
- We design a paired, multi-source evaluation protocol that holds a fixed panel of scheduling policies constant across independently sourced workload windows, operating regions, and metrics, so that ranking-portability statistics are computed on genuinely paired observations rather than on separately tuned, per-source experiments.
- We assemble a mechanism-diverse, fixed scheduler panel spanning classical, fairness-aware, KV-aware, deadline-aware, and faithfully reimplemented external policies, selected under an outcome-independent selection rule fixed before any pilot result was observed (§5).
- We define an uncertainty and reversal-detection methodology that distinguishes practically meaningful rank reversals from microscopic sign changes, using preregistered practical-margin and confidence-interval criteria, and that treats every completion-conditioned metric’s undefined cases explicitly rather than by imputation (§5).

- We define a benchmark sample-complexity methodology that asks how many independent workload windows are required before a reported ranking is likely to reproduce, rather than assuming a conventionally sized window set is sufficient.
- We characterize the workload corpus underlying the benchmark, establishing that the workload sources used are measurably different in observed descriptor distributions and strongly separable under a leakage-resistant, chronologically blocked evaluation (§4).
- We plan a paired public artifact release of the benchmark’s frozen workload identities, policy panel, protocol, and (once generated) outcome and analysis artifacts, to support independent reproduction (§11).

Consistent with the scientific state of the project at the time of writing (§3), this manuscript does not claim that scheduler rankings are stable, that they are unstable, that any reversal exists, or that any scheduler in the panel wins. Sections 6 through 8 state the exact, preregistered result that each pending experiment will report, without asserting its direction in advance.

2 Related Work

2.1 Serving Architectures and Scheduling Mechanisms

A substantial body of systems work has improved LLM inference throughput and latency through scheduling and memory-management mechanisms. Orca introduced iteration-level scheduling, interleaving requests at the granularity of individual decoding steps rather than whole requests [24]. vLLM introduced PagedAttention, a paged virtual-memory-style KV-cache allocator that substantially raised achievable batch sizes and became a widely adopted serving-system reference point [8]. Sarathi-Serve introduced chunked prefill, splitting long prefill computations across multiple decode-interleaved steps to reduce head-of-line blocking under continuous batching [1]. Splitwise and DistServe independently proposed disaggregating the compute-bound prefill phase and the memory-bound decode phase onto separate hardware pools connected by fast interconnects, each showing throughput or cost improvements from phase-specific resource allocation [9, 27]; Mooncake extends this direction with a KV-cache-centric disaggregated architecture that treats a shared, elastic KV cache – rather than the compute engines themselves – as the central resource to schedule around [10]. Llumnix adds dynamic, cross-instance request migration and rebalancing on top of a serving cluster [17]. DynamoLLM addresses cluster-level energy and cost efficiency by dynamically reconfiguring an inference cluster’s resource allocation under latency SLOs [18].

A second cluster of work targets scheduling *policy* design more specifically. VTC (Virtual Token Counter) defines an explicit token-level fairness objective for continuous-batching schedulers and proves a bound on the resulting service-difference guarantee [16]. S^3 predicts each request’s output length ahead of generation and uses that prediction to pack KV-cache memory more tightly at admission time, trading occasional misprediction-driven preemption for higher achievable batch size [7]. Fu et al. [6] instead learns a pairwise ranking model over waiting requests to prioritize decode order – a distinct mechanism from S^3 ’s admission-time length prediction, despite both papers appearing at NeurIPS in consecutive years and both targeting throughput under uncertain output length. FastServe, JITServe, PARS, Libra, SLAI/RAD, SCORPIO, LMetric, Apt-Serve, and ProServe extend this space further with iteration-level preemption, SLO-aware admission under imprecise request information, pairwise learning-to-rank variants, dynamic request partitioning, theoretically motivated multi-objective scheduling, heterogeneous-SLO-aware admission, lightweight

scheduling metrics, adaptive hybrid-cache scheduling, and unified multi-priority scheduling, respectively [4, 11, 12, 14, 19, 20, 22, 25, 26]. Each of these papers reports an improvement over one or more baselines on the workload(s) chosen for its own experiments; none of them, individually or collectively, asks whether the *relative ordering* they report would be preserved under an independently sourced workload, a different metric, or a different operating load – the question this benchmark is built around (§2.6).

2.2 Production Workloads and Characterization

A parallel literature documents that LLM-serving workloads observed in production are themselves heterogeneous. BurstGPT releases 10.31 million traces collected from a regional deployment of Microsoft’s Azure OpenAI GPT service over 213 days, characterizing request burstiness, conversation structure, and response-length behavior, and is now published as a full KDD 2025 paper rather than only as a preprint [21]. Two official Microsoft releases document Azure’s own internal LLM-inference traffic: Splitwise analyzes production conversation and code-completion traces collected on Azure in November 2023 [9], and DynamoLLM analyzes a longer, one-week Azure trace collected in May 2024 [18]. anonymized per venue record; see primary source [3] characterizes KV-cache reuse patterns at a large cloud provider, finding that reuse probability and lifespan are workload-category-dependent in ways that synthetic workload studies had not captured. see primary source for full author list [15] reports a one-year longitudinal characterization of production traffic spanning workload evolution, caching behavior, and load balancing (a recent preprint at the time of writing; §2.6). ServeGen instead characterizes a worldwide production inference service in order to build a principled per-client workload *generator*, motivated explicitly by the position that no single fixed trace is representative enough on its own [23]; because it characterizes the same underlying Alibaba Cloud platform as this project’s Bailian/Qwen source, we do not count it as an independent workload provider (§4). TraceLab characterizes a distinct workload family entirely – coding-agent sessions rather than single-turn or short-multi-turn chat requests – and is used here only as out-of-distribution context, not as one of this project’s primary Pilot-V2 sources.

2.3 Simulators and Benchmark Frameworks

Vidur is the closest existing infrastructure precedent to LSSP: a large-scale simulation framework purpose-built for exploring LLM-inference system configurations at a fidelity and speed unavailable from running full experiments on real hardware [2]. LLMServingSim and its successor LLMServingSim 2.0 provide a related but architecturally different hardware/software co-simulation infrastructure, adding iteration-granularity simulation with algorithmic-redundancy reuse and, in the 2.0 revision, explicit support for heterogeneous accelerators and disaggregated serving techniques [5, 13]. These simulators are built to let researchers explore serving-system design points cheaply; none of them is built around, or reports, a benchmark of *comparative scheduler ranking portability* across independently sourced workloads in the sense this paper defines. LSSP’s own simulator (§5) is project-local infrastructure, not a contribution positioned against this literature; we make no claim that its output is equivalent to production hardware latency or throughput, only that it supports a fixed, controlled benchmark protocol for relative comparisons (§5).

2.4 Workload Dependence and Benchmark Sensitivity

Prior work has already established, individually, that LLM-serving workloads are heterogeneous (§2.2) and that individual serving mechanisms and schedulers are sensitive to batching structure, prefill/decode balance, load level, and service-level objectives (§2.1). We treat both of these as

accepted background, not as a novel contribution of this work. What we do not find in the literature reviewed here is a direct study of whether a *comparative ordering* among a fixed panel of schedulers is itself portable across independently sourced workloads, operating regions, and metrics – the distinction this paper’s Introduction develops between single-system sensitivity and comparative-ranking portability.

2.5 Ranking Portability and Benchmark Methodology

The statistical methodology this benchmark uses – Kendall’s tau-b and Spearman’s rho for pair-wise ranking agreement, paired comparison and block-bootstrap confidence intervals for ranking-uncertainty quantification, and a probability-of-correct-selection framing for benchmark sample complexity – draws on standard nonparametric statistics and comparative-algorithm-ranking methodology (§5); these are general statistical tools, not literature claims specific to LLM serving, and are cited here only as the methods this benchmark’s analysis plan applies to its own, independently generated data.

2.6 Position of LSSP

LSSP asks a different question than the majority of the prior work surveyed above. Prior work generally studies a scheduler, a serving system, a simulation environment, or a workload generator/characterization in its own right. LSSP instead studies the portability of *comparative* scheduler rankings across independently sourced workload families, operating regimes, and metrics, using a fixed, mechanism-diverse scheduler panel and a preregistered protocol. To our knowledge, the LSSP benchmark is distinct in treating the portability of comparative scheduler rankings itself as the primary object of study across independent workload sources, operating regimes, and metrics, while explicitly quantifying ranking uncertainty and benchmark sample complexity. Vidur is the closest infrastructure precedent (§2.3); LLMservingSim and its successor are related but differently scoped simulator infrastructure; the recent production-characterization literature (§2.2) establishes the workload heterogeneity this benchmark’s design responds to, but does not itself study cross-source ranking portability. This wording is intentionally narrower than an unconditional claim of priority; we do not claim to be the first to study LLM-serving workload dependence or scheduler sensitivity, only that we did not identify a prior benchmark treating cross-source, cross-metric, cross-load-region scheduler-ranking portability as its primary object of study under the primary-source review conducted for this manuscript.

A note on provider independence is warranted here because it bears directly on how several of the sources above should be read together (§4 gives the full treatment). BurstGPT’s own reported methodology collects its traces from a regional customer’s use of Microsoft’s Azure OpenAI GPT service [21]; this project’s own Azure-sourced traces are Microsoft’s own official releases of internal Azure LLM-inference-service traffic [9, 18]. Both therefore sit on the same underlying cloud provider’s infrastructure, even though they are collected by different teams through different mechanisms and released as distinct, independently versioned artifacts. We describe BurstGPT and this project’s Azure traces as distinct, independently released workload sources throughout this manuscript, and we do not describe them as independent providers.

3 Study Design

3.1 Research Questions

We reproduce the project’s frozen research questions verbatim (see `docs/RESEARCH_QUESTIONS.md` for the full changelog):

- RQ1.** How stable are scheduler rankings across independent workload sources?
- RQ2.** How stable are scheduler rankings under temporal, provider, and domain shifts?
- RQ3.** To what extent do rankings obtained on synthetic stress workloads transfer to rankings on independent real-trace-derived workloads?
- RQ4.** Which source-native observable workload characteristics are associated with cross-distribution scheduler rank reversals?
- RQ5.** How many independent workload windows are required before a comparative scheduler ranking becomes statistically stable?
- RQ6.** Do representative simulated scheduler rankings and cross-workload rank reversals reproduce on a real serving engine?

Load-level dependence is a secondary robustness/sensitivity analysis applied to RQ1–RQ4’s findings, not a headline research question (§3.3).

3.2 Study Overview

LSSP evaluates a fixed panel of scheduling policies (§5.1) against a frozen corpus of workload windows drawn from independently sourced traces (§4.1), across a calibrated grid of operating regions (§5.5) and a fixed set of primary and completion-conditioned metrics (§5.6). Every policy is run against every window under every operating region, producing a matrix of cells from which per-source, per-region, and per-metric scheduler rankings are derived. Ranking-portability statistics (Kendall’s tau-b, Spearman’s rho, top- k overlap, and pairwise reversal detection) are then computed across pairs of sources, regions, and metrics, using workload window as the statistical unit (§3.4). This design was frozen (§3.5) before any comparative scheduler outcome existed.

3.3 Evidence Independence / Publication Boundaries

This project shares a lineage and a simulator codebase with two other, separately scoped manuscripts, and its design was explicitly audited for overlap against both before this study’s protocol was frozen (`docs/OVERLAP_LEDGER.md`, `docs/GO_NO_GO_GATES.md` Gate A, `docs/EVIDENCE_INDEPENDENCE_PLAN.md`):

- **“The Exploitability Gap in LLM-Serving Scheduler Portfolios” (LLM 2026).** That manuscript’s `public_replay_load_scaling_v1/v2` experiment used a fixed set of 60 canonical public-trace windows (20 BurstGPT + 20 Azure-2023 conversation + 20 Azure-2023 code), a load-factor grid $\{1, 2, 4, 8, 16, 32, 64, 128\}$, an 8-policy portfolio, and ANWG/SBS-VBS exploitability diagnostics. That result is treated here as `PRIOR_RESULT_REFERENCE_ONLY`: citable as prior motivation, never restated as this project’s own evidence. This project’s own window sampling is independently constructed (§4.3) and verified non-identical to that 60-window set; load-level dependence is accordingly demoted to a secondary analysis (§3.1) using this project’s own calibration protocol (§5.5), never that load grid.

- **“Module Counterfactuals for Attributing LLM Serving Scheduler Effects” (a separate module-intervention manuscript).** This study makes no module-counterfactual or module-attribution claim; RQ4’s workload-descriptor analysis (§5) is an offline, descriptive association analysis, not a deployed selector or an intervention-based attribution method.
- Other project lineage (an unrelated database/systems research thread) is unconnected to this manuscript’s content and is not referenced here.

3.4 Experimental Unit and Paired Design

The statistical unit of analysis is the workload window: a fixed-size (200-request) contiguous slice of a source trace, uniquely identified and frozen before any scheduler outcome existed (§4.3). Every policy in the panel (§5.1) is run against every window under every calibrated operating region (§5.5), so that ranking-portability statistics are computed on genuinely paired observations – the same window, region, and metric definition, varying only the policy – rather than on separately tuned per-source experiments. Source, operating region, and metric are treated as factors of interest; policy identity is the object whose relative ordering is measured.

3.5 Preregistration

The workload window set (120 windows: 40 per source, 10 Stage-0-reused + 30 Pilot-V2-new per source; §4.3), the scheduler panel and its selection rule (§5.1), the metric contract (§5.6), the six-region operating-load calibration (§5.5), and the ranking-portability analysis plan (reversal-detection thresholds, bootstrap procedure, sample-complexity design) were each frozen, and cross-checked against a structural validation gate, before any Pilot-V2 comparative scheduler outcome was generated. `docs/RANKING_PORTABILITY_PILOT_V2_PROTOCOL.md`, `docs/RANKING_PORTABILITY_ANALYSIS_PLAN.md`, and `docs/RANKING_PORTABILITY_WINDOW_FREEZE.md` record the exact frozen content and the gate checks each passed.

4 Workload Corpus and Characterization

4.1 Sources

LSSP’s primary Pilot-V2 corpus draws from three independently released workload sources: BurstGPT [21], Microsoft’s official Azure 2024 LLM-inference trace [18], and a Bailian/Qwen production trace release. Azure’s 2023 conversation/code traces [9] and Stage-0’s historical windows contribute additional context but are not part of the primary Pilot-V2 132-window matrix beyond the 10-per-source Stage-0-reused windows described below. TraceLab [28] is retained as an out-of-distribution, coding-agent workload family for future expansion, not as one of the three primary sources, because its independence from a prior 512-window public sweep cannot be fully verified from public artifacts alone (`docs/TRACELAB_PROVENANCE_RESOLUTION.md`).

As established in §2.6, BurstGPT’s own reported collection methodology logs traffic from a regional customer of Microsoft’s Azure OpenAI GPT service, and this project’s Azure trace is Microsoft’s own official release of internal Azure LLM-inference-service traffic. Both therefore sit on the same underlying cloud provider’s infrastructure. We describe BurstGPT and Azure as distinct, independently released workload sources – different collection methodology, different customer population, different release provenance and version history – and we do not describe them as independent providers. Bailian/Qwen (Alibaba Cloud) is a genuinely different underlying platform from both.

4.2 Representation and Provenance

Each source is ingested through a dedicated, field-level-audited adapter (`docs/DATA_FIELD_PROVENANCE.md`) that maps the source’s native schema onto a common internal request representation (arrival time, input tokens, output tokens, total tokens, session identifier where available, model class). No field is fabricated where a source’s native schema lacks it (for example, BurstGPT’s documented schema has no tenant, KV-reuse, SLO, or failure-code field, and none is synthesized). License status for each source is recorded in `docs/DATA_LICENSE_AUDIT.md` (BurstGPT: CC-BY-4.0) and re-verified before any confirmatory use of a real, full-scale sample.

4.3 Window Construction

The Pilot-V2 corpus freezes exactly 120 windows of 200 requests each: 40 per source (`azure_llm_2024`, `bailian_qwen`, `burstgpt`), each composed of 10 windows reused verbatim from the Stage-0 pilot (§5.4) plus 30 new windows drawn by a deterministic, outcome-blind, source-symmetric pinned-extension sampler that uses only valid-row index ranges and fixed seeds – never scheduler metrics, policy identities, or Stage-0 tie/non-tie outcomes (`BURSTGPT_OUTCOME_TARGETING = NO`, structurally verified). Windows are assigned to EARLY/MIDDLE/LATE chronology strata by deterministic relative chronology (60/30/30 windows respectively); Stage-0-reused windows are all EARLY. The freeze passed a full structural validation gate – unique window identities, exact per-source and Stage-0/new counts, fixed `request_count = 200`, no prohibited overlap, valid chronology, and absence of any scheduler/policy/load-calibration outcome field in the selection rule – recorded with a canonical content hash (excluding generation timestamp; full value in Table 1) before any Pilot-V2 load calibration or scheduler outcome was generated.

4.4 Descriptors

Each window is described by a 28-feature descriptor vector spanning prompt- and output-token-length statistics (mean, p90, long-prompt fraction at the 512/2048/8192-token thresholds), arrival-timing statistics (mean arrival rate, inter-arrival percentiles), and derived load-pressure proxies. Three deterministically redundant pressure-proxy features ($R^2 = 1.0$ against other included features) are retained in the underlying dataset but excluded from the classifier input used in §4.6. Prompt-token figures compose differently across sources for architectural reasons: TraceLab’s and Azure’s prompt-token fields are potentially multi-turn-cumulative context, while BurstGPT’s is a single per-request figure. This is a property of how each source’s underlying workload is architected (a growing agent/conversation session versus a single stateless API call), not an adapter artifact, and it is preserved as an explicit caveat on every prompt-length-based claim below.

4.5 Cross-Source Differences

Under a leave-one-time-bucket-out (EARLY/MIDDLE/LATE), grouped, chronologically blocked cross-validation over $n = 400$ windows (window size 200), a random forest classifier distinguishes the four descriptor-labeled source classes with pooled balanced accuracy 1.000 (mean $\pm$ std across three folds: 1.000 ± 0.000); a multinomial logistic regression reaches 0.988 balanced accuracy (0.987 ± 0.007 across folds); and a depth-limited (depth ≤ 3) decision tree reaches 0.980 balanced accuracy (0.980 ± 0.018 across folds, worst single fold 0.955 with LATE held out). This grouped, chronologically blocked evaluation supersedes an earlier random-split result (which agreed at balanced accuracy 1.0) for any manuscript claim, because random splitting is vulnerable to within-window-adjacent-time

leakage that the grouped evaluation is structurally immune to (regression tested; `tests/test_separability_pipeline_no_leakage.py`).

4.6 Source Separability

We report the final characterization result as `SEPARABILITY_RESULT_CONFIRMED_WITH_CAVEATS`. The safe claim supported by this evidence is that the workload sources’ observed descriptor distributions are measurably different, and that source identity is strongly separable under a leakage-resistant, chronologically blocked evaluation. This is not evidence that the underlying providers’ conversations or tasks intrinsically differ in length or structure, only that their *observed, request-level* descriptor figures do, for a mixture of architectural and possibly-genuine-content reasons that this classifier-based analysis cannot separate (§4.4).

4.7 Drivers of Differences

An attribution analysis triangulating held-out permutation importance, single-feature balanced accuracy, leave-one-feature-out deltas, and feature-group-ablation deltas (all under the same grouped chronological cross-validation) identifies two tiers of driving features. Tier 1 – incrementally load-bearing, with nonzero permutation importance and a measurable accuracy cost when the `prompt_length` feature group is removed (-0.0051) – consists of `long_prompt_fraction_512`, `long_prompt_fraction_2048`, and `prompt_tokens_mean`. Tier 2 – individually strong single-feature classifiers (balanced accuracy up to 0.980) whose removal, individually or as a whole feature group, costs 0.000 accuracy because their signal is redundant with Tier 1 – includes total/prompt-token percentile features, inter-arrival percentiles, and mean arrival rate. Prompt-length structure, specifically the fraction of long prompts at the 512- and 2048-token thresholds and mean prompt length, is therefore the primary, non-redundant driver of source separability; arrival-timing structure and output length are individually predictive but carry no additional signal once prompt-length features are present. A critical-ablation check confirms this separability is broad rather than dependent on any single feature: removing the two strongest Tier-1 features together changes random-forest balanced accuracy by at most -0.0025 .

4.8 Window-Size Sensitivity

The Pilot-V2 corpus uses a fixed window size of 200 requests throughout, matching Stage-0 and the LLM 2026 comparison window size for methodological consistency. A dedicated window-size sensitivity sweep (varying window size and re-checking separability/discriminability) has not yet been run; this is noted as an open item in §10 rather than a result reported here.

5 Scheduler Benchmark Methodology

5.1 Scheduler Panel

The PRIMARY panel comprises 11 policies, executed alongside 2 additional `STYLE_APPROXIMATION` policies used only for a mandatory robustness check (13 policies executed in total): a classical arrival-order baseline (`fifo`, also the load-calibration reference policy), a deadline-aware policy (`edf`), a slack/urgency-aware policy (`least_laxity_first`), a service-length-aware policy (`estimated_service_time_first`), a fairness-aware policy (`weighted_fair_share`), a KV/memory-pressure-aware policy (`kv_constrained_online`), an admission-control/SLO-guard

policy (`admission_control`), and faithful reimplementations of four external mechanisms: vLLM-style token-budget continuous batching in both its pre-chunked-prefill and chunked-prefill forms (`vllm_faithful`, `vllm_chunked_prefill_faithful`), Sarathi-style chunked prefill (`sarathi_faithful`), and an SLO-aware scheduling policy in the SLAI/RAD family (`slai_faithful`). The two `STYLE_APPROXIMATION` robustness policies are a token-budget-proxy style approximation (`vllm_style_token_budget`) and an SLO-guard-proxy style approximation (`scorpio_style_slo_guard`); two further faithful reimplementations requiring disaggregated/cross-instance simulator semantics (`distserve_faithful`, `llumnix_faithful`) are executed only in a secondary stratum, never pooled with the single-GPU colocated PRIMARY leaderboard, and one candidate (`apt_serve_faithful`) is excluded entirely as `scaffolding_only` – not a validated reimplementation at the time of this freeze.

A policy enters the PRIMARY panel only if, at the time it was added to the comparability audit (frozen the day before Stage 0 executed), it satisfies an outcome-independent selection rule: its mechanism predates this protocol’s freeze, it occupies a mechanism family not already saturated by an existing panel member, it has adequate implementation fidelity and passes its own unit tests, it is simulator-semantically compatible with the single-GPU colocated execution model, and it is independently reproducible from a cited reference. No policy was added to or removed from the panel in response to any Stage-0 outcome (§5.4); the panel document predates Stage-0 execution by a full day, and mechanism families identified as under-implemented (preemptive/multilevel scheduling in the FastServe sense, JITServe, LMetric) were deliberately left as documented future-work candidates rather than fast-tracked into the panel after seeing Stage-0’s BurstGPT collapse.

5.2 Fidelity Taxonomy

Every executed policy is labeled with one of five fidelity classes: `FAITHFUL_EXTERNAL` (a validated reimplementation of a specific published mechanism), `REPOSITORY_NATIVE_CLASSICAL` (a standard scheduling discipline implemented natively in this simulator, not claimed as a reproduction of any specific paper), `SIMULATOR_PROXY` (a mechanism-motivated policy without a single canonical external paper), `STYLE_APPROXIMATION` (an approximate, style-level proxy for a published mechanism, explicitly excluded from the PRIMARY headline panel and used only for robustness checks), and `scaffolding_only` (an unaudited, not-yet-validated implementation, excluded entirely). We do not claim that any `STYLE_APPROXIMATION` or `SIMULATOR_PROXY` policy is an official or validated reimplementation of its namesake published system.

5.3 Execution Model

Each (window, policy, operating-region) cell is executed as a fully deterministic discrete-event simulation: given identical inputs, the simulator produces bit-identical outputs, with no consumed randomness (verified by a repeated-run determinism test). This determinism is what permits the paired design of §3.4: two policies’ outcomes on the same window and region differ only because of the policy’s own scheduling decisions, not because of independent sampling noise.

5.4 Stage-0 Discriminability Pilot (Historical)

Before the Pilot-V2 protocol described in the remainder of this section was frozen, a preregistered discriminability pilot (“Stage 0”) executed a real, 1,080-cell matrix (array 1213964; three sources, three load regions, a six-policy PRIMARY subset) to check, before committing to the full protocol, that the panel would yield non-trivial scheduler differentiation under a fixed load protocol without post-hoc cherry-picking. **This is historical pilot evidence, not the confirmatory Pilot-V2**

result, and it never measured cross-source ranking portability – that question was and remains explicitly out of Stage-0’s scope. Stage 0’s own five-criterion discriminability analyzer returned four passing criteria (77.8% of (source, window, load-region) triples non-tied, 86.7% of windows non-tied, no universal policy collapse, and 100% of non-tied cells showing meaningful metric variation) and one genuine failure: BurstGPT contributed only 14.3% of non-tied conditions, against a 15% floor, versus $\sim 42.9\%$ each for the other two sources – a result reported as `STAGE0_NO_GO`, robust to the completion-conditioned-metric zero-completion convention used. A follow-up diagnostic traced this to a mechanism-level explanation: differentiation was strongly source-dependent, attributable to a near-total collapse of five of the six Stage-0 policies into pairwise bit-identical outcomes on BurstGPT’s structurally short-prompt, near-constant-output traffic – not to BurstGPT receiving weaker or fewer calibrated load conditions than the other sources. This motivated the Pilot-V2 protocol’s larger, mechanism-diverse 13-policy panel and denser six-region load grid (§5.5), applied symmetrically to every source, never to BurstGPT alone. Detailed Stage-0 mechanism analysis is deferred to an appendix; it is not restated here as a ranking-portability finding.

5.5 Operating-Region Calibration

A FIFO-only calibration pass establishes six operating regions – `LOW` (load factor 0.5), `PRE_KNEE` (0.8), `KNEE` (1.0), `POST_KNEE` (1.1), `OVERLOAD` (1.2), and `HIGH_PRESSURE` (1.5) – across all 120 frozen windows (720 calibration cells total; 720 valid, deterministic, 0 failed/missing/ duplicated). We report this as calibration provenance, not a comparative scheduler result: FIFO is the calibration reference policy, and the calibration’s role is to fix load-factor-to-region mappings before any other policy is executed against them, not to characterize FIFO’s relative performance. As a calibration characteristic worth preserving, 301 of the 720 calibration-region records (93 BurstGPT, 101 Bailian/Qwen, 107 Azure) recorded zero completions under FIFO at that region’s load factor; this is a property of the calibration design at those specific (window, region) combinations, the frozen calibration validity gate passed regardless (`PHASE11_CALIBRATION_VALID = YES`), completion-conditioned metrics are simply undefined (not zero) in those records (§5.6), and no calibration redesign was performed after observing this count.

5.6 Metrics

Three metrics are always defined for every completed cell: completion fraction, arrival-normalized weighted goodput (ANWG, the primary metric), and weighted completion fraction. A second group of metrics is defined only conditional on at least one request completing: SLO-violation rate, weighted goodput, mean/p95/p99 latency, and request/token throughput are each `NaN` if and only if completion fraction is exactly zero for that cell. Mean/p95 time-to-first-token is subject to a stricter, separately checked precondition (at least one completed request recorded a first-token time), because TTFT can be undefined even when completion fraction is positive. No undefined conditional-metric value is ever imputed as 0 or 1, in either direction, and never after inspecting which policy or source it would affect. Where a policy’s value is undefined for a given (window, region, metric) condition, that policy is excluded from the ranking-derived statistic (Kendall’s tau-b, Spearman’s rho, top- k overlap, pairwise reversal detection) for that specific condition and metric only – never scored as a loss or a tie, and never causing the whole condition to be dropped for policies that did complete. Every ranking-derived table reports its exclusion count (`n_conditions_excluded_for_undefined_metric`) alongside the statistic itself, since a source-concentrated exclusion rate is itself a reportable finding under RQ4, not a footnote.

5.7 Paired Design and Repetitions

Because the simulator is fully deterministic (§5.3), a single execution per (window, policy, region) cell is sufficient; ranking-uncertainty quantification (§6) instead comes from resampling over the independent workload windows themselves (block bootstrap), not from repeated stochastic runs of the same cell.

5.8 Mechanism Telemetry

Every executed cell additionally records a seven-field mechanism-activation telemetry summary: queue-depth (mean/max), batch-saturation (mean/max fraction of `max_active_sequences` in use), prefill/decode contention fraction, KV-occupancy (mean/max fraction of `max_kv_tokens` in use), admission-control activation count, preemption/reorder event count, and token-budget saturation fraction. Each field is always defined: a value of zero means either that the corresponding mechanism is genuinely absent for that policy (for example, preemption count is exactly zero for every policy that never issues a preempt/swap/migrate action, which is most of the panel) or that the underlying simulator mode does not implement a separate prefill/decode phase or token-budget model – not an instrumentation failure. This telemetry is collected for every cell but is not used to support any mechanism-level finding in this manuscript; RQ4’s telemetry-to-reversal association analysis (§6) is pre-specified and will only be run once Pilot-V2 outcomes exist.

5.9 Robustness Analyses

Every headline ranking-portability result (§6) is pre-specified to be checked against a fixed set of robustness analyses: the 11-policy high-fidelity subset (PRIMARY panel excluding both `STYLE_APPROXIMATION` policies), leave-one-source-out re-estimation, window-size sensitivity, an alternate completion-conditioned-metric convention, the full six-region load grid, a temporal-split sensitivity check, and mechanism-family-exclusion checks. These robustness analyses are part of the frozen protocol; their results are reported alongside §6 once Pilot-V2 executes (§??).

6 Scheduler Ranking Portability

Every subsection below is a result contract: an exact specification of what will be reported once Pilot-V2 executes (§11 states current execution status), reproduced from `docs/MANUSCRIPT_RESULT_CONTRACT.md`, the project’s single source of truth for contract wording. No subsection below asserts a direction of finding. Either a high-portability or a low-portability outcome is a complete, reportable result under this design (§9).

6.1 Discriminability

[PENDING PILOT-V2 RESULT: fraction of non-tied (window, load-region) conditions per source, per metric, over the full 13-policy / 6-region Pilot-V2 matrix, with Wilson 95% confidence intervals per source (method reused from the Stage-0 diagnostic, §5.4). Historical Stage-0 evidence (3 sources $\times$ 3 regions $\times$ 6-policy subset only) is reported separately in §5.4 and is not substituted for this result.]

6.2 Cross-Source Rankings

[PENDING PILOT-V2 RESULT: Kendall’s tau-b and Spearman’s rho between each pair of the three sources’ policy rankings, per metric (§5.6), per operating region, with block-bootstrap 95% confidence intervals over windows. Reported alongside top-1/top-3 overlap and `n_conditions_excluded_for_undefined_metric` per cell. No claim of “stable” or “unstable” rankings is made before this table exists.]

6.3 Pairwise Reversals

[PENDING PILOT-V2 RESULT: every (policy A, policy B) pair’s reversal frequency and count of practically meaningful reversals only – defined as exceeding the preregistered 10% practical-margin threshold with a confidence interval excluding zero – reported separately from microscopic sign changes. No reversal count mixing the two categories is reported.]

6.4 Temporal/OOD Portability

[PENDING PILOT-V2 RESULT: three separately labeled comparisons – within-BurstGPT EARLY/MIDDLE/LATE tau, Azure-2024 own-window calendar-split tau, and cross-source (provider/domain) tau from §6.2 – never merged into a single “temporal/domain OOD” number.]

6.5 Load Conditioning

Load-level dependence is a secondary robustness/sensitivity analysis (§3.1), not a headline claim of cross-source rank instability by itself.

[PENDING PILOT-V2 RESULT: tau and reversal frequency across the six frozen operating regions (§5.5), per source, reported as a robustness/sensitivity table using this project’s own independently calibrated load protocol – never LLM 2026’s {1, 2, 4, 8, 16, 32, 64, 128} grid or its 60-window set (§3.3).]

6.6 Metric Conditioning

[PENDING PILOT-V2 RESULT: tau between every pair of metrics in the metric contract (§5.6), per source, reported alongside whether any exclusion-driven gap from undefined completion-conditioned metrics is concentrated in one source – a candidate explanatory factor if so, reported descriptively, not causally.]

6.7 Robustness

[PENDING PILOT-V2 RESULT: sensitivity of §6.1 through §6.6’s headline statistics to the 11-policy high-fidelity subset, leave-one-source-out re-estimation, window-size choice, the alternate completion-conditioned-metric convention, the full six-region load grid, a temporal-split sensitivity check, and mechanism-family-exclusion checks (§5.9).]

7 Benchmark Sample Complexity

[PENDING RESULT: probability of recovering the full $n = 40$ -window ranking at $n \in \{5, 10, 20, 30, 40\}$ windows per source, using at least 500 draws per n , per source and per metric, including a comparison between a concentrated (single-source-heavy) and a distributed (evenly-spread)

window budget. We report the smallest n at which recovery probability first exceeds the preregistered 0.9 threshold; no specific value of n is claimed before this analysis exists.]

This analysis directly answers RQ5 (§3.1) and depends on the full Pilot-V2 outcome matrix (§6); it is not run on Stage-0’s historical, smaller pilot (§5.4).

8 Real-System Validation

[PENDING RESULT: sign agreement, Kendall’s tau, and reversal-agreement between simulated and real-vLLM policy rankings, for a frozen, objectively selected pair of validation cases – the largest-effect-size reversal identified in §6.3 plus one stable-ordering control condition. Case selection must be frozen from this study’s own §6.2/§6.3 results before any real-vLLM run is executed, and cannot be written until those results exist.]

This analysis answers RQ6 (§3.1). We do not claim, here or anywhere in this manuscript, that simulated absolute latency or throughput values match real-hardware values; the only claim in scope is agreement in *relative ranking* and *reversal direction* between the simulator and a real vLLM deployment on the selected cases (docs/REAL_SYSTEM_VALIDATION_PLAN.md).

9 Discussion

Because §6–§8 report no comparative outcome yet, we structure this discussion around both directions the pending result may take, so that the discussion below requires no rewriting once the outcome is known – only the removal of whichever branch did not occur.

9.1 If Ranking Portability Is High

If scheduler rankings prove largely portable across the three independently sourced workload families, operating regions, and metrics tested (§6.2–§6.6), the practical implication is that future scheduler comparisons may not need to be repeated across many independently sourced workloads to be trustworthy: a smaller, well-chosen evaluation footprint could suffice, reducing the evidence burden of scheduler evaluation. A high-portability result would also constitute evidence of a kind of robustness for the panel’s mechanisms themselves – that their relative strengths and weaknesses are not artifacts of any one workload’s specific structure. At the same time, a high-portability finding on this benchmark’s three sources and fixed operating-region grid would not license an unconditional claim of portability to every workload a practitioner might encounter; the boundary conditions under which portability was observed (source identity, load range, metric set, and simulator fidelity; §10) would need to be stated precisely, and §7’s result would still bound how much evidence is required before a specific portability claim is statistically warranted.

9.2 If Ranking Portability Is Low

If scheduler rankings instead prove fragile – with practically meaningful, statistically supported reversals concentrated in particular sources, regions, or metrics (§6.3) – the practical implication is the opposite: single-workload scheduler evaluations risk over-generalizing, and a comparative claim of the form “scheduler A outperforms scheduler B ” should be read as conditional on the workload, metric, and load region it was measured under, unless independently re-verified. This would motivate multi-source evaluation as a de facto requirement for general-purpose scheduler comparisons going forward, and would sharpen the importance of §7’s sample-complexity result for practitioners who cannot afford to test every candidate workload distribution. A low-portability

finding would also raise the practical question of which workload characteristics predict a reversal (RQ4, §4.5, §5.8) – explanatory, not selector-building, in scope (§3.3).

9.3 Mechanism Equivalence and Collapse

Independent of which branch above applies at the ranking-portability level, the mechanism telemetry collected for every cell (§5.8) allows a finer-grained discussion of *why* rankings behave as observed – for example, whether policies that appear to tie on a given workload do so because their scheduling decisions genuinely converge (as Stage 0’s historical diagnostic found for several policies on BurstGPT’s structurally short-prompt traffic; §5.4) or because the metric used cannot distinguish otherwise-different decision sequences. This discussion is deferred to the post-Pilot-V2 revision of this section, once §6.2–§6.6 exist to anchor it.

9.4 Workload Descriptor Interpretation

The workload-separability result (§4.6) establishes that the three sources are observably different, and that prompt-length structure is the primary driver of that separability (§4.7). Whatever RQ4’s descriptor-to-reversal association analysis finds, it should be read against this backdrop: an association between a descriptor and a reversal does not by itself establish that the descriptor is a genuine *causal* driver of the reversal, only that it is a source-separating feature that co-occurs with where reversals were or were not observed.

9.5 Benchmark Construction and Practical Reproducibility

Regardless of outcome direction, this study’s paired design, fixed scheduler panel, and preregistered protocol (§3) are intended to make the benchmark itself reusable: a future scheduler design can be dropped into the same panel and evaluated against the same frozen windows, operating regions, and metric contract, using the planned public artifact (§11) rather than re-deriving the protocol from this manuscript’s prose.

10 Threats to Validity

Simulator abstraction. All results in this manuscript are produced by a discrete-event simulator, not a real serving engine (§5.3). We do not claim that simulated absolute latency or throughput values match real-hardware values; only relative ranking and reversal direction are claimed to transfer, and only for the frozen validation cases in §8, once that section’s result exists.

Fidelity of reimplemented policies. FAITHFUL_EXTERNAL policies are validated reimplementations, not the original authors’ code; residual implementation gaps relative to the original published systems are possible despite unit-test validation (§5.2).

STYLE_APPROXIMATION baselines. The two style-level approximation policies are explicitly excluded from the PRIMARY headline panel and used only for a robustness check; they must not be read as official or validated reimplementations of their namesake systems (§5.2).

Three primary workload families. The Pilot-V2 corpus draws from three primary sources (§4.1); a fourth candidate source (TraceLab) is retained only as out-of-distribution context because its independence from a prior public window sweep cannot be fully verified. Any portability finding in §6 is scoped to these three sources plus whatever out-of-distribution evidence TraceLab can responsibly contribute, not to LLM-serving workloads in general.

Dataset/provider dependence. As established in §4.1, BurstGPT and this project’s Azure traces are not established to come from independent underlying providers – both trace back to Microsoft Azure infrastructure, collected through different mechanisms. Any cross-source finding involving BurstGPT and Azure should be read with this in mind; Bailian/Qwen remains a genuinely independent platform.

Prompt-token semantic differences. Prompt-token fields compose differently across sources (single-request figure for BurstGPT versus potentially multi-turn-cumulative context for Azure and TraceLab; §4.4). Any prompt-length-based claim in this manuscript carries this caveat and should not be read as evidence that the sources’ underlying conversations differ in length, only that their observed, request-level prompt-token figures do.

Unresolved historical byte-level overlap. An exact, byte-level reconstruction of a prior related project’s specific 60-window public-trace replay set could not be independently re-derived from this project’s available checkout; that project’s own recorded parameters and final verdict are taken on the strength of direct manuscript inspection, not independently re-derived from local artifacts. This project’s own window construction is independently verified as non-identical (§4.3, §3.3), but the historical byte-level reconstruction gap itself remains an open item.

Homogeneous hardware / execution assumptions. The simulator executes every cell under a fixed, single-GPU-located execution model for the PRIMARY panel; two disaggregation-requiring policies are confined to a secondary stratum and never pooled with the PRIMARY leaderboard (§5.1).

Completion-conditioned undefined metrics. Several metrics are undefined, not zero, when a cell records zero completions (§5.6); this is a real methodological choice with a real cost – an undefined-metric policy is excluded from that specific condition’s ranking-derived statistics, which lowers effective sample size for those conditions and must be read alongside every ranking-derived table’s reported exclusion count.

Phase-11 zero-completion prevalence. 301 of 720 FIFO-only calibration-region records recorded zero completions (§5.5). This is disclosed as a calibration characteristic, not a headline finding; it does, however, mean that a non-trivial share of (window, region) combinations will contribute no completion-conditioned-metric observation to §6’s downstream analyses for any policy at that specific region, which is a structural constraint on statistical power in the highest-pressure regions in particular.

Scope of real-system validation. §8’s validation is limited by design to a small, frozen set of case-selected conditions (largest-effect-size reversal plus one stable-ordering control), not a full re-execution of the Pilot-V2 matrix on real hardware; agreement or disagreement on those specific cases should not be over-generalized to every cell in §6.

11 Reproducibility and the LSSP Artifact

We plan to release the LLM-Serving Scheduler Portability Benchmark (LSSP) as a public artifact at the Hugging Face repository slug `SoroshVahidi/llm-serving-scheduler-portability`. **As of this manuscript, that artifact is planned but not yet released.** The intended release will include: the frozen workload window identities and their provenance (§4.3); the scheduler panel and its configuration (§5.1); the six-region calibration protocol and its outputs (§5.5); the full set of paired policy outcomes across windows, regions, and metrics, once generated (§6); the mechanism telemetry collected for every cell (§5.8); the ranking-portability analysis outputs (Kendall/Spearman statistics, reversal tables, block-bootstrap confidence intervals); and the reproduction scripts used to regenerate every table and figure in this manuscript from those frozen inputs.

Every generated artifact in the planned release is designed to carry, or link via a provenance manifest, its generating repository commit SHA, resolved experiment-config hash, workload-manifest hash, policy-registry version, dataset-source checksums, environment version, and (where relevant) per-cell seed, so that any released row can be regenerated from this project’s own commit history alone rather than depending on an external, possibly-rewritten historical commit.

The immutable scientific identifiers already fixed at the time of this manuscript are recorded in Table 1 and in `docs/ARTIFACT_HASH_LEDGER.md`: the Phase-10 workload-window freeze content hash, the Phase-10 compact index hash, the Phase-11 prelaunch freeze contract hash, the Phase-11 raw FIFO calibration output hash, and the Phase-11 region-assignment output hash. These five identifiers were independently re-verified as unchanged, and consistent across every cross-referencing project document, immediately before this manuscript’s authoritative branch was committed (§12).

12 Conclusion

We have formulated the portability of a comparative scheduler ranking, $R(W, M, L)$, as a benchmark object distinct from both workload characterization and single-scheduler evaluation, and we have designed, frozen, and documented a paired, multi-source evaluation protocol – a mechanism-diverse fixed scheduler panel, a calibrated six-region operating grid, an explicit metric contract, and an uncertainty/reversal-detection methodology – before generating any comparative outcome. We have established, as project-local evidence available now, that the three primary workload sources underlying this benchmark are measurably different and strongly separable in their observed descriptor distributions (§4), and, as historical pilot evidence, that scheduler differentiation under this project’s protocol is real but strongly source-dependent, motivating the larger panel and denser load grid used going forward (§5.4). The benchmark’s central questions – whether comparative scheduler rankings are portable across independent sources, operating regions, and metrics, how many workload windows are required before such a ranking is trustworthy, and whether simulated rankings reproduce on a real serving engine – remain open at the time of this manuscript and are stated as explicit result contracts (§6–§8) rather than asserted in either direction. We plan to release LSSP as a public, independently reproducible artifact (§11) once these results exist.

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Table 1: Immutable scientific identifiers, re-verified unchanged immediately before this manuscript’s authoritative branch was committed (`docs/ARTIFACT_HASH_LEDGER.md`).

Artifact	SHA-256
Phase-10 scientific window (content hash)	0d1aa06ccbee352207327ea369ae75f12e91c0cda006c813a41b381effd29eef
Phase-10 compact index	d78ec1087fedae02174ca093a9860c70468be336ccb1d7e6de756c81ba331e53
Phase-11 prelaunch freeze contract	e2564ea9484190832de50f63173c4b73ae054d6ae7008bb4ff6648c8dc917f7b
Phase-11 raw FIFO calibration	201caaf04476ad8737ef6079fc0d6cb4e864601711d0b96c88750a717d8b2a6a
Phase-11 region-assignment output	9fcb92f9ea1206ce185194527ada35d0e3b91bf4904be7ae23ba9ea997c17574

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